

CORPORATE FINANCE - WILDWOOD CASE STUDY

1. How much business (operating) risk does Wildwood face?

Business risk refers to the variability of operating income (EBIT) caused by the firm's industry environment, its cost structure, and competitive conditions. In the Wildwood case, the company's business risk is relatively low. The snack-food sector is mature and stable, sales have grown steadily, and EBIT has increased consistently with only temporary disruption during the 2007–08 crisis. This intense rivalry limits pricing power, meaning Wildwood must rely on volume growth and efficiency rather than price increases to drive profit. Wildwood emphasizes operational efficiency and strict control as a “way of life”, avoids high-risk product bets, and operates in a category with habitual demand. By maintaining an efficient production and distribution, Wildwood reduces variability in its operating profits. Additionally, management pursues cautious, incremental growth, focusing on extensions of existing products and small acquisitions rather than risky ventures.

Overall, Wildwood exhibits low to moderate operating risk, below many industrial firms and comparable to peers like PepsiCo.

Another aspect affecting Wildwood's risk profile is its short-term financial policy, which is very flexible. The firm holds large cash balances, cash was ~18% of total assets at the end of 2011, and has zero debt, indicating a preference for liquidity and safety.

From a quantitative point of view, we can assess operational risk by applying Modigliani's second proposition, making some assumptions and using the following formula:

$$WACC = (1 - T_c)r_{debt} \left(\frac{D}{D + E} \right) + r_{equity} \left(\frac{E}{D + E} \right)$$

Since the debt is zero, what emerges is that WACC equals equity. However, we also know that:

$$r_{equity} = r_{assets} + \frac{D}{E}(r_{assets} - r_{debt})$$

Therefore:

$$WACC = r_{equity} = r_{assets}$$

Ra is the fundamental value that isolates operational risk, regardless of its financial structure. Since all capital comes from shareholders, the return required by equity holders (i.e., ROE) coincides with the return required on all the company's assets. In the absence of debt, there is no difference between the return generated by assets and the return required by shareholders.

So, we can state that:

$$r_{assets} = r_{equity} = ROE = 12,5\%$$

This indicates a moderate risk profile, consistent with that of a consumer goods company with relatively stable margins. This rate also represents the minimum return required by shareholders, as there is no financial risk arising from leverage. In summary, Wildwood's

business risk is relatively low for a consumer products company: the firm operates in slow-growing industry and employs cautious management practices that limit earnings volatility.

2. How much financial risk does Wildwood have today (pre-recapitalization) and how much financial risk would the company face at each of the three alternative debt-to-capital ratios presented in case Exhibit 4?

Before recapitalization, Wildwood carries no debt, so its financial risk is zero: there is no possibility of default, no interest burden, and no leverage effect on shareholder returns. Financial risk is evaluated primarily using debt-to-capital ratios, interest coverage, bond ratings, and the cost of debt.

At 20% debt-to-capital, the firm's debt would be \$145M, with extremely high interest coverage (36.9), an implied AAA/AA rating, and a low borrowing rate of 2.85%. Financial risk at this level is low, with leverage increasing ROE but adding almost no default risk.

At 40% debt-to-capital, debt rises to \$290M and interest coverage remains strong (11.82). The firm would likely be rated BBB and borrow at around 4.4%. Financial risk becomes moderate, but still at investment-grade and very manageable.

At 60% debt-to-capital, debt reaches \$435M and coverage drops to 4.52. The implied rating falls to the B range and the cost of debt increases to 7.7%. Financial risk at this level is high, with default exposure and strong sensitivity of ROE to leverage: the firm is at junk-level credit quality.

Overall, financial risk increases non-linearly with leverage, moving from negligible (0% debt) to low (20%), moderate (40%), and high (60%).

We can reach the same conclusions by considering the case quantitatively.

To understand the impact of debt and calculate the financial risk we can apply the following formula:

$$(r_{assets} - r_{debt}) \frac{D}{E}$$

Since Wildwood currently carries no debt ($D=0$), the firm is fully equity-financed. This means that the firm's financial risk today is zero. This ultra-conservative balance sheet means that shareholders' earnings (EPS) and returns (ROE) fluctuate only with operating performance, not with any leverage effects. Indeed, Wildwood's pre-recapitalization ROE of ~12.5% in 2011 is entirely driven by its business results and large equity base. However, as analysts note, the absence of debt also depresses return on equity with no leverage to amplify profits. Wildwood's ROE and EPS are lower than they could be.

SCENARIO	Ra	Rd	D	E	FINANCIAL RISK
20% Debt-to-capital	12,5%	2,85%	145	580	2,41%

40% Debt-to-capital	12,5%	4,40%	290	435	5,40%
60% Debt-to-capital	12,5%	7,70%	435	290	7,20%

As the level of debt increases, the firm's equity becomes riskier because debt holders have prior claims on cash flows. Investors demand a higher return (R_e) as compensation, even though R_a stays fixed at 12.5%. The difference between R_e and R_a quantifies the incremental financial risk introduced by leverage.

These findings confirm the logic of MM Proposition II: financial leverage increases the risk borne by equity holders, which results in a higher cost of equity. At 60% debt, the financial risk premium reaches 7.2%, signaling substantial shareholder exposure to fixed financial obligations.

This analysis provides a framework to assess the trade-offs in capital structure: while moderate debt may enhance shareholder returns, excessive leverage leads to increased risk and cost of equity, potentially offsetting the benefits.

3. How much value could Wildwood create for its shareholders at each of the three alternative debt levels? (hint, use the company financials to make some calculations)

To estimate how much value Wildwood could create for its shareholders under each of the proposed recapitalization scenarios, we compute the present value of the interest tax shield associated with taking on debt. In order to calculate it, we need to multiply the corporate tax rate times the amount of debt raised.

Wildwood faces a corporate tax rate of 35.5 percent. At a 20 percent debt-to-capital ratio, the company would take on \$145 million in debt. The value of the tax shield generated by this level of leverage is therefore $0.355 \times 145 = \$51,5$ million. At a 40 percent debt-to-capital ratio, the firm would raise \$290 million in debt. The associated tax shield is $0.355 \times 290 = \$103,0$ million. Finally, at a 60 percent debt-to-capital ratio, Wildwood would borrow \$435 million, creating a tax shield worth $0.355 \times 435 = \$154,4$ million.

These calculations show that the value created through the tax shield increases in a linear way as the firm increases its leverage. However, while the tax benefit rises proportionally with debt, the firm's cost of debt and financial risk do not increase linearly. In fact, the cost of borrowing accelerates at higher leverage levels as the company's credit rating deteriorates. Thus, although the 60 percent structure generates the largest nominal tax benefit, it also adds higher financial risk and potential distress costs that must be weighed against the incremental value created.

From the data provided, we can also observe that Wildwood buys back its shares on the market, thus reducing the number of outstanding shares. This improves the ratios per share. In fact, EPS grows substantially from \$2.88 without debt to \$3.19 with 20% debt, \$3.31 with 40% debt and finally decreases slightly to \$3.11 with 60% debt. The performance of EPS underlines that

the financial leverage within 40% increases the equity per share, thanks also to the lower number of shares. However, with more than 40% of debt, the trend reverses, as interest on debt erodes net profit.

Analysing the ROE, it increases in correspondence with debt. The higher the percentage of debt, the greater the increase in ROE. This is because part of the capital is replaced by debt which is cheaper.

Finally, by calculating the P/E it is possible to assess the value of the shares perceived by the market and therefore how much value the shareholders actually receive. The results of the calculation give a P/E of 14.47 with no debt, 15.02 with 20% debt, 15.11 with 40% debt, and 16.75 with 60% debt. So, the P/Es are all increasing compared to the initial value without debt. This shows that P/E increases consistently with the increase in EPS. It should also be noted that the company is increasing its premium over the initial market price of \$41.67. In particular, in the first scenario with debt at 20% and a premium on the buyback price at +15%, the market perceives an improvement in profitability per share. The same goes for the P/E with 40% debt, calculated with a maximum EPS of \$3.31. In the third scenario, a declining EPS and an increasing repurchase price compared to the other scenarios must be considered. In fact, there is a high P/E compared to the previous two. The increase in P/E in the third scenario is due to the increase in the repurchase price and the higher debt risk.

In conclusion, the first two scenarios create shareholder value by improving EPS with low risk. In the third scenario, however, EPS decreases due to the cost of debt, but the share repurchase price increases. So, P/E is increasing but not because the company is creating value.

4. What debt-to-capital structure would you recommend as optimal for Wildwood Cola & Snacks? Justify your choice.

Based on the financial implications, strategic considerations, and risk profile presented in the case, the optimal capital structure for Wildwood is a 40 percent debt-to-capital ratio, since it balances value creation with financial stability.

The 40 percent structure generates a substantial tax shield of approximately \$103 million, which is almost double the value created at 20 percent debt, yet does not expose the company to the severe financial risk associated with the 60 percent alternative. At 40 percent leverage, Wildwood maintains an investment-grade credit rating (BBB), supported by a very strong interest-coverage ratio of 11.82, which indicates that the firm can meet its debt obligations even under adverse conditions. The cost of debt at this leverage level remains reasonable at 4.4 percent.

In contrast, a 60 percent debt-to-capital structure would move the company into speculative-grade territory with a B rating, raising the cost of borrowing to 7.7 percent and significantly increasing the risk of financial distress. Although the 60 percent structure produces a larger tax shield, the additional \$51 million of value created comes at the expense of a huge decline in

credit quality, an increase in distress costs, and higher vulnerability to earnings volatility, so the trade-off is not worth it.

Furthermore, the 40 percent structure is consistent with Wildwood's culture, which the case describes as conservative, disciplined, and oriented toward operational stability. The company has always avoided taking on debt, and its management emphasizes predictable performance and controlled risk. Moving directly to 60% leverage wouldn't be in line with the company's financial conservatism.

Finally, this level of leverage Wildwood matches common practices among competitors in the snack and beverage industry. It also improves overall performance without pushing the firm into a high-risk credit category.

5. What are the advantages of adding debt to the capital structure?

Introducing debt into Wildwood's capital structure offers several important financial and strategic advantages. The most direct benefit of using debt is the creation of interest tax shields. Because interest expenses are tax-deductible, taking on leverage reduces the company's taxable income and therefore its tax liability. As the case notes, Wildwood faces a corporate income tax rate of 35.5%. Applying this tax rate to any amount of issued debt immediately generates value for shareholders by increasing after-tax cash flows. This mechanism explains why each of the proposed recapitalization scenarios creates value, ranging from \$51.5 million at 20% to \$154.4 million at 60% debt.

A second advantage of adding debt is the increase in ROE thanks to leverage. In the case's Exhibit 4, the firm's ROE and EPS increase as debt levels rise because equity is being reduced through the share repurchase while earnings remain supported by a stable operating business. For example, the EPS rises from 2.88 to 3.31 under the 40% recapitalization scenario. This illustrates how moderate leverage can improve equity performance and shareholder returns.

A third advantage is that debt financing can discipline managerial behaviour by reducing excess free cash flow. As described in the case, Wildwood has historically been conservative, accumulating cash on the balance sheet. The case also notes the company's "obsession with cost control" and high operating discipline. Introducing debt obligations (interest and principal payments) can reinforce this discipline by requiring management to prioritize efficient capital allocation and avoid wasteful spending.

A fourth advantage is that leveraging the capital structure allows Wildwood to optimize its capital costs and better align with industry norms. The case shows that many comparable firms in the snack and beverage sector operate with moderate leverage, including Snyder's-Lance (23% D/Cap) and PepsiCo (approximately 50% D/Cap). By introducing debt, Wildwood reduces its overreliance on equity and moves toward a more efficient capital structure relative to competitors, potentially improving investor perception and lowering the firm's WACC.

A fifth advantage is that debt is a cheap source of financing since it gives a pre-defined and low-risk return to investors. In fact, bond holders receive fixed interest payments, have

priority over equity holders, and also can claim assets in case of bankruptcy. These advantages give the firm the possibility to offer lower interest rates and so pay less costs on debt.

Finally, taking on debt provides strategic flexibility and creates an opportunity for shareholder distribution. The case indicates that the proceeds of the new debt could be used to repurchase a significant portion of the outstanding equity, rewarding shareholders directly, improving share liquidity and enhancing the value of remaining equity shares.

6. How would issuing debt impact the company's taxes and expected costs of financial distress?

Issuing debt would have two opposing effects on Wildwood's financial position: it would reduce the company's taxes through the creation of interest tax shields, while simultaneously increasing its expected costs of financial distress as leverage rises. Both effects are central to understanding the trade-offs involved in recapitalizing the firm.

First, adding debt has a direct impact on taxes because interest payments are tax-deductible. Every dollar of interest the company pays reduces its taxable income by that amount, lowering the total tax burden. This tax benefit is substantial: under the three recapitalization scenarios provided in the case, the interest tax shields create \$51.5 millions of value at a 20% debt ratio, \$103.0 million at 40% debt, and \$154.4 million at 60% debt. These figures highlight that the tax savings grow proportionally with the amount of debt issued.

At the same time, issuing debt also increases expected costs of financial distress, which grow disproportionately as leverage rises. These costs are not explicitly quantified in the case and in general they are difficult to calculate. These expenses include direct costs, such as legal and administrative fees, higher interest rates demanded by bondholders, and a reduced incentive to rely on debt financing. In addition, there are indirect costs, such as asset devaluation resulting from suboptimal managerial decisions driven by concerns over the firm's condition, declines in sales, operational disruptions, turnover among variable employees, and lower overall employee morale.

However, it is possible to build estimated costs of financial distress considering the data given in the case and making some assumptions. The expenses can be obtained by using this formula:

$$E(\text{CFD}) = P(\text{Distress}) * \text{Cost of Distress}$$

The probability value for the three scenarios can be found looking at one of the most relevant costs, which is the interest rate that creditors demand as Wildwood's credit risk increases. This premium represents the market's assessment of the expected costs. Given every interest rate and the credit ratings of the scenarios, the probability of distress can be deducted as follows:

DEBT-TO-CAPITAL RATIO	ASSUMED BOND RATING	INTEREST RATE	P(DISTRESS)
20%	AAA/AA	2,85%	0,05%

40%	BBB	4,40%	0,20%
60%	B	7,70%	3%

The bankruptcy costs (cost of distress) are in average around 15% of the market capitalization value of the unlevered firm, so given that:

$$V_u = \text{market cap} - \text{excess cash}$$

$$V_u = (\text{outstanding shares} \times \text{stock price at the end of 2011}) - \text{excess cash}$$

DEBT-TO-CAPITAL RATIO	P(D)	C(D)	E(CFD)
20%	0,05%	2117,85	1,058925
40%	0,2%	2117,85	4,2357
60%	3%	2117,85	63,5355

Looking at the table, it is evident that the expected costs of financial distress rise as the debt-to-capital ratio increases. This highlights a trade-off between higher leverage and the value of the company's tax shield.

7. How would the financial markets (stock price) react if the company increased its financial leverage?

The financial markets would likely respond positively to a moderate increase in Wildwood's financial leverage, but the size of the reaction would depend on the level of debt the company chooses to adopt. A recapitalization that raises leverage to an efficient but still safe level would be expected to increase the firm's stock price, while excessive leverage could provoke a negative market reaction due to higher financial risk. The following table illustrates the increase in price of the shares of the company as the debt-to-capital ratio grows.

DEBT TO CAPITAL	PREVIOUS STOCK VALUE	PRICE PER SHARE REPURCHASE	DIFFERENCE	PERCENTAGE INCREASE
20%	41,67	47,92	6,25	15%
40%	41,67	50	8,33	20%
60%	41,67	52,09	10,42	25%

If Wildwood were to increase its leverage to the 20 percent or 40 percent debt-to-capital structures proposed in the case, the market response would almost certainly be favourable. By adding debt, the company would generate substantial tax shields, which translate directly into increased firm value. Because the stock price reflects the market's assessment of firm value, these increases would be expected to raise the share price proportionally. Additionally, the recapitalization involves repurchasing shares, which reduces the number of shares outstanding. This reduction of the equity base, combined with the increase in firm value, would amplify the impact of the recapitalization on the stock price. The case data confirm this effect: EPS rises from 2.88 before the recapitalization to 3.15 under 20 percent leverage and to 3.31 under 40 percent leverage, reflecting increased equity value per share.

However, the market's reaction would be different if Wildwood pursued the 60 percent debt-to-capital structure. While this structure produces the largest tax benefit (\$154.4 million), the case shows that it results in a drop in the company's credit rating from investment-grade BBB at 40 percent leverage to speculative-grade B at 60 percent. The cost of debt rises sharply to 7.7 percent, and the interest coverage ratio falls to 4.52, exposing the firm to higher financial risk and causing growing concerns among shareholders.

We can therefore conclude that moderate leverage would raise the firm's stock price, while extreme leverage would diminish it.

8. How could Wildwood implement a more aggressive capital structure?

Wildwood can implement a more aggressive capital structure through several financing actions that shift the company away from its current zero-debt policy and toward a mix of debt and equity. The most direct method is the one illustrated in the case exhibits: issuing long-term debt and using the proceeds and a portion of the firm's excess cash to repurchase outstanding shares. This immediately reduces the equity base, raises the debt-to-capital ratio, and enhances important values such as EPS, ROE, and the interest tax shield.

Another way to aggressively increase leverage would be to issue debt specifically to fund a large special dividend, which achieves the same reduction in equity as a buyback but avoids the rising repurchase premiums that occur as the repurchase size grows.

Wildwood could also adopt a policy in which future capital investments, acquisitions, and capacity expansions are financed with debt rather than with retained earnings. This gradually increases leverage over time while still preserving the company's ability to internally fund operations.

The company could make use of bank financing by taking on a revolving credit facility to support working capital needs, reducing cash balances, and increasing the proportion of debt in the capital structure.

Finally, selling non-operating assets and using the proceeds to buy back outstanding shares or to fund special dividends would further reduce equity and raise leverage. Any of these strategies would allow Wildwood to shift from an all-equity structure to a more leveraged one.

9. What is the drawback to the method of recapitulation proposed in the case? What alternative methods (in addition to the option proposed in the case) could be used to increase leverage and decrease equity?

The recapitalization method proposed in the case, which is issuing long-term debt and using the proceeds and \$55 million of excess cash to repurchase common shares, has a key drawback: the repurchase becomes more and more expensive as its size grows, since the company will have to pay progressively higher premiums above the current market price to buy its own shares.

These rising premiums diminish the net value created for shareholders and erode part of the tax benefits from increasing leverage. The approach also consumes a substantial portion of the company's cash reserves, which contradicts Wildwood's preference for liquidity and financial safety, and it introduces a sudden increase in financial risk by adding fixed interest obligations that must be met even in weaker business periods.

In addition, a large and immediate buyback can send ambiguous signals to the market. It can raise concerns about whether the firm has limited investment opportunities or whether management believes the stock is mispriced. For these reasons, although the method is effective in increasing leverage quickly, it is not necessarily the most efficient approach.

Some alternative methods could achieve the same increase in leverage without the specific drawback of expensive repurchases. For example, issuing debt to fund a large special dividend reduces equity immediately but avoids paying a premium over market price.

The company could also shift its financing policy so that future capital expenditures, plant expansions, and potential acquisitions are financed with debt rather than retained earnings, allowing leverage to rise over time in a more sustainable way.

If management seeks a smoother transition, in compliance with the culture, the firm could issue hybrid securities, such as convertible bonds, which add leverage at lower interest costs and with lower perceived risk, while postponing potential dilution.

These alternatives offer a more flexible and cheap way to create a more efficient capital structure while avoiding the rising repurchase premiums.

10. Considering Wildwood's corporate culture, what arguments could you use to persuade CEO Jennings or his successor to adopt and implement your recommendation? That is, make the case to the CEO (pros and cons) for your recommendation.

To persuade CEO Jennings or his successor to adopt a more leveraged capital structure, we must acknowledge Wildwood's risk averse culture while showing that leverage could increase shareholder value, aligning with the company's values, without compromising the firm's stability.

The core argument should show that adding a reasonable amount of debt would help achieve the company's objectives rather than go against them, as an increase in leverage would generate tax savings through the tax shield.

Since Wildwood currently holds a large cash balance that is earning almost no return, substituting part of its idle equity with low-cost debt would increase ROE while also providing economic benefits. Moreover, the firm operates in a mature and stable industry with predictable cash flows and relatively low capital expenditure requirements, making it an ideal candidate to carry moderate leverage safely.

At the same time, it is important to emphasize that adding leverage does not require abandoning the financial conservatism that the CEO values. The firm is not being asked to adopt a highly aggressive or risky structure, but rather to move from an extreme policy, which is having zero debt, to a more efficient and balanced one that maintains high safety margins. It is possible to choose a debt level that allows the company to remain above the investment grade rating, doesn't deteriorate the interest coverage ratio and the bankruptcy risk. The company can preserve its culture of discipline by selecting debt instruments with long maturities, low interest obligations, and great flexibility. In fact, having a targeted level of leverage can enhance financial flexibility, because it opens access to capital markets, prevents cash from accumulating and avoids market pressure.

The CEO needs to also keep into account the expectations of the investor community, as they view the absence of debt in the company as a barrier to achieving higher returns, particularly since the competitors in the market are able to operate with leverage and still have high rating profiles. Addressing the shareholders' concerns and increasing leverage could result in higher stock prices.

One of the drawbacks to these arguments could be that the addition of leverage introduces fixed obligations and a new concern for a management team that is used to having a very low financial risk. However, we know that we can adopt a mild approach, avoiding excessive one-time recapitalizations and selecting a safe capital structure.

To conclude, moderate leverage supported by stable cash flows represents a logical step for Wildwood to comply with their value of maximizing shareholder value and creating long-term sustainability, rather than representing a divergence with Jennings's views.

11. Discuss the efficiency of Wildwood's current capital structure, according to theory discussed in class.

According to corporate finance theory, Wildwood's current capital structure is financially conservative to an extreme degree and therefore inefficient in creating value.

In efficient markets, the cost of debt is lower than the cost of equity, since the debt holders face lower risk than shareholders, so by choosing not to use any debt financing, the company renounces to the tax benefits provided by the tax shield, since interest payments are deductible and directly reduce taxable income. This means the company pays more in taxes than necessary, leaving less cash available for shareholders and reducing the firm's enterprise value. Moreover, because the company finances all assets with equity, its ROE is lowered, due to the large equity base and large cash holdings that dilute profitability.

From the perspective of the trade-off theory, Wildwood is positioned on the “under-leveraged” side of the curve. The firm’s business risk is moderate and stable, since sales and margins are predictable, the product portfolio is diversified, and the firm has a long history of consistent performance, suggesting that the probability of financial distress from moderate leverage is very low. In such circumstances, theory predicts that the optimal capital structure should include some level of debt, where the marginal benefits of the tax shield still exceed the expected costs of distress. We can conclude that Wildwood operates far below the theoretical optimum, sacrificing tax savings and financial efficiency.

The pecking-order theory also points toward inefficiency. Under this framework, firms prefer internal financing first, then safe debt, and only as a last resort the issuance of new equity.

Relying exclusively on equity, Wildwood is inconsistent with the theory’s prediction that firms with stable cash flows should at least incorporate some debt into their capital structure, both to prevent dilution and avoid accumulating idle cash. The latter represents an opportunity cost and reduces capital efficiency. Holding such a large liquidity buffer may increase perceived safety, however it also signals an overly restrictive financial policy and a failure to optimize capital use, both of which contradict the notion of shareholder value maximization emphasized by the CEO.

Finally, comparing Wildwood with its competitors, what really catches the eye is the fact that PepsiCo and Snyder’s-Lance use from 20% to 50% of leverage, while maintaining a strong credit profile and high rating. The markets would expect Jennings’ company to comply with the industry’s standards.

12. Assuming profitable operations and retention of earnings, how long would the company remain at its optimal debt level after a one-time debt issue?

If Wildwood issues debt only once and then continues operating profitably while retaining a meaningful portion of its earnings, the company will naturally drift away from its optimal capital structure rather quickly. This occurs because retained earnings increase the equity base each year, while the one-time debt issue remains fixed unless the firm actively manages its leverage. As equity grows and debt stays constant, the debt-to-capital ratio declines steadily, causing the firm to move back toward its under-leveraged position. The precise speed of this decline depends on Wildwood’s profitability, payout ratio, and internal growth rate, but given the company’s high cash generation and historically conservative dividend policy, the erosion of the initial debt ratio would be relatively fast.

Corporate finance theory shows that when a firm generates positive net income and retains a portion of it, equity increases automatically through retained earnings, and for a company like Wildwood these retained earnings accumulate quickly. In effect, the firm’s internal growth rate acts as a dilution mechanism on leverage: the stronger the growth in equity, the faster the capital structure shifts back toward the unlevered position. Without continuous leverage adjustments, the firm would only remain near its optimal debt level for a short period of time before the debt ratio drops below the target.